

EX:	
DATE:	

AIM:

ALGORITHM:

CODE:

```
import pandas as pd
```

```
df=pd.read_csv(r"C:\Users\Keertu\Downloads\tourism_dataset_5000.csv")
```

```
print(df)
```

	Tourist ID	Age	Interests \
0	1	48	['Architecture', 'Art', 'History']
1	2	37	['Cultural', 'Nature']
2	3	43	['History', 'Art', 'Architecture']
3	4	46	['Cultural', 'Art', 'Architecture']
4	5	53	['Architecture', 'Art']
...	...	...	...
4995	4996	31	['Architecture', 'History', 'Cultural']
4996	4997	28	['Architecture', 'Cultural', 'Nature']
4997	4998	42	['History', 'Cultural', 'Nature']
4998	4999	58	['Cultural', 'Art', 'History']
4999	5000	32	['Nature']

	Preferred Tour Duration	Accessibility	Site Name \
0	5	False	Eiffel Tower
1	6	False	Colosseum
2	6	True	Machu Picchu
3	8	False	Colosseum
4	5	True	Colosseum
...	...	...	...
4995	3	False	Great Wall of China
4996	7	True	Great Wall of China
4997	3	True	Machu Picchu
4998	5	False	Colosseum
4999	4	False	Colosseum

	Sites Visited	Tour Duration \
0	['Eiffel Tower', 'Great Wall of China', 'Taj M...']	7
1	['Great Wall of China']	1

	Sites Visited	Tour Duration \
0	['Eiffel Tower', 'Great Wall of China', 'Taj M...']	7
1	['Great Wall of China']	1
2	['Eiffel Tower']	2
3	['Machu Picchu', 'Taj Mahal']	5
4	['Machu Picchu', 'Taj Mahal', 'Great Wall of C...']	7
...	...	...
4995	['Great Wall of China']	2
4996	['Colosseum']	1
4997	['Taj Mahal']	1
4998	['Taj Mahal', 'Great Wall of China', 'Colosseum']	7
4999	['Great Wall of China']	1

	Route ID	Tourist Rating	System Response Time	Recommendation Accuracy \
0	1000	1.6	3.73	97
1	2000	2.6	2.89	90
2	3000	1.7	2.22	94
3	4000	2.0	2.34	92
4	5000	3.7	2.00	96
...	...	...	...	...
4995	4996000	3.9	2.53	100
4996	4997000	4.3	1.53	100
4997	4998000	3.2	3.99	89
4998	4999000	2.5	2.13	98
4999	5000000	1.9	2.99	92

	VR Experience Quality	Satisfaction
0	4.5	3
1	4.5	3
2	4.7	3
3	4.7	3
4	4.8	4
...	...	...
4995	4.4	4
4996	5.0	4
4997	4.4	3
4998	4.4	3
4999	4.8	3

[5000 rows x 14 columns]

print(df.head())

	Tourist ID	Age	Interests \
0	1	48	['Architecture', 'Art', 'History']
1	2	37	['Cultural', 'Nature']
2	3	43	['History', 'Art', 'Architecture']
3	4	46	['Cultural', 'Art', 'Architecture']
4	5	53	['Architecture', 'Art']

	Preferred Tour Duration	Accessibility	Site Name \
0	5	False	Eiffel Tower
1	6	False	Colosseum
2	6	True	Machu Picchu
3	8	False	Colosseum
4	5	True	Colosseum

	Sites Visited	Tour Duration	Route ID \
0	['Eiffel Tower', 'Great Wall of China', 'Taj M...]	7	1000
1	['Great Wall of China']	1	2000
2	['Eiffel Tower']	2	3000
3	['Machu Picchu', 'Taj Mahal']	5	4000
4	['Machu Picchu', 'Taj Mahal', 'Great Wall of C...']	7	5000

	Tourist Rating	System Response Time	Recommendation Accuracy \
0	1.6	3.73	97
1	2.6	2.89	90
2	1.7	2.22	94
3	2.0	2.34	92
4	3.7	2.00	96

	VR Experience Quality	Satisfaction
0	4.5	3
1	4.5	3
2	4.7	3
3	4.7	3
4	4.8	4

print(df.info())

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 14 columns):
#   Column                                Non-Null Count  Dtype
---  ---                                -
0   Tourist ID                            5000 non-null   int64
1   Age                                    5000 non-null   int64
2   Interests                             5000 non-null   object
3   Preferred Tour Duration               5000 non-null   int64
4   Accessibility                         5000 non-null   bool
5   Site Name                             5000 non-null   object
6   Sites Visited                         5000 non-null   object
7   Tour Duration                         5000 non-null   int64
8   Route ID                              5000 non-null   int64
9   Tourist Rating                       5000 non-null   float64
10  System Response Time                  5000 non-null   float64
11  Recommendation Accuracy               5000 non-null   int64
12  VR Experience Quality                 5000 non-null   float64
13  Satisfaction                         5000 non-null   int64
dtypes: bool(1), float64(3), int64(7), object(3)
memory usage: 512.8+ KB
None

```

```
print(df.describe())
```

	Tourist ID	Age	Preferred Tour Duration	Tour Duration \
count	5000.000000	5000.000000	5000.000000	5000.000000
mean	2500.500000	43.886000	5.52280	3.97720
std	1443.520003	15.268858	1.71186	1.99967
min	1.000000	18.000000	3.00000	1.00000
25%	1250.750000	30.000000	4.00000	2.00000
50%	2500.500000	44.000000	6.00000	4.00000
75%	3750.250000	57.000000	7.00000	5.00000
max	5000.000000	70.000000	8.00000	9.00000

	Route ID	Tourist Rating	System Response Time \
count	5.000000e+03	5000.000000	5000.000000
mean	2.500500e+06	3.002740	3.238802
std	1.443520e+06	1.164353	1.016123
min	1.000000e+03	1.000000	1.500000
25%	1.250750e+06	2.000000	2.360000
50%	2.500500e+06	3.000000	3.230000
75%	3.750250e+06	4.000000	4.130000
max	5.000000e+06	5.000000	5.000000

	Recommendation Accuracy	VR Experience Quality	Satisfaction
count	5000.000000	5000.000000	5000.000000
mean	92.492200	4.493160	3.534200
std	4.616081	0.290596	0.736029
min	85.000000	4.000000	3.000000
25%	88.000000	4.200000	3.000000
50%	92.000000	4.500000	3.000000
75%	97.000000	4.700000	4.000000
max	100.000000	5.000000	5.000000

```
import sqlite3

conn=sqlite3.connect("student.db")

cursor=conn.cursor()

cursor.execute("""

create table if not exists students(

id integer primary key,

name text,

age integer,

department text

)

""")

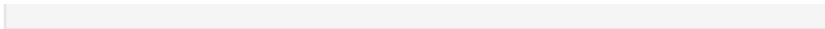
cursor.execute("insert into students(name,age,department) values('keertu',19,'CSBS')")

cursor.execute("insert into students(name,age,department) values('shabu',28,'Architecture')")

conn.commit()

conn.close()

print("database created sucessfully")
```



```
database created sucessfully
```

```
import sqlite3

import pandas as pd

conn=sqlite3.connect("student.db")

query="select *from students"

df=pd.read_sql(query,conn)

print(df)
```

	id	name	age	department
0	None	keertu	19	CSBS
1	None	shabu	28	Architecture
2	None	keertu	19	CSBS
3	None	shabu	28	Architecture

```
data={  
    "Name":["keerthu","shabu","dabu"],  
    "Age":[19,28,30],  
    "department":["CSBS","Architecture","Engineer"],  
    "Id":[1,2,3]  
}
```

```
df=pd.DataFrame(data)
```

```
df.to_excel("output.xlsx",index=False)
```

```
print("Excel File Exported")
```

Excel File Exported

```
df.to_json("output.json",orient="records",indent=4)
```

```
print("JSON file exported")
```

JSON file exported

```
df.to_sql("students",conn,if_exists="replace",index=False)
```

```
print("sql exported")
```

sql exported

RESULT: